Candidate's Examination No.....

MARKING SCHEME

SECTION A (15 Marks)

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
A	D	B	D	B	B	C	C	C	D

@ 01 mark ⇒ 10 Marks

2.

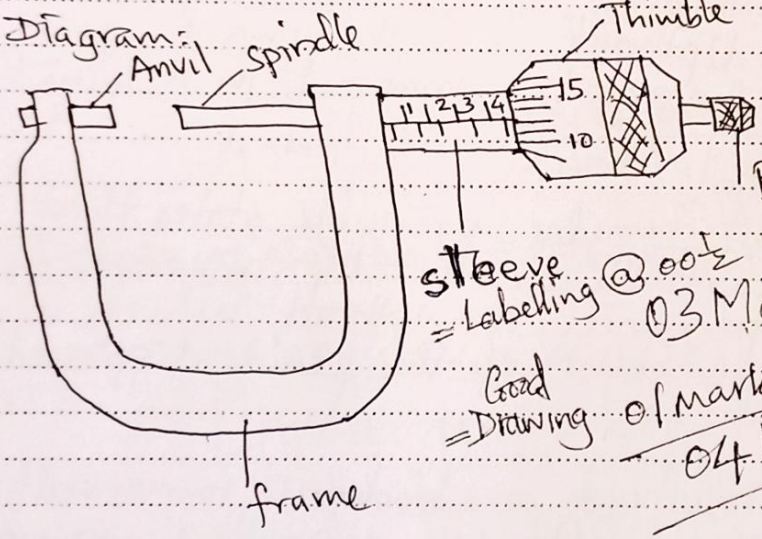
i	ii	iii	iv	v
B	A	D	E	C

@ 01 mark ⇒ 05 Marks

SECTION B. [70 Marks]

3. (a)(i) By using a micrometer screw gauge. 01 mark

(ii) Diagram:



= Labelling @ 00½ 03 Marks

= Good Drawing 01 mark

04 Marks

(b). i. Force of gravity 00½ mark
ii. Electromagnetic force 00½ mark
iii. Strong nuclear force 00½ mark
iv. Weak nuclear force 00½ mark

(c). A laboratory should be well ventilated and its door should open outwards.

(ii). It should be fitted with fire extinguisher in accessible position
(iii). Laboratory floors should not be polished as it will make them slippery.
iv). An adequate equiped First Aid kit should be placed in every laboratory.
(v) cabinets and drawers should be included in the design of a laboratory so as to be used for storing apparatus.

ANY 3 @ 01 mark 03 Marks

[TOTAL 10 marks]

4 (a) (i) Real weight - is the weight of a body/an object in air or the exactly weight of a body in air. 01 mark

(ii) Upthrust - is the upward force exerted on an object Immersed in a fluid 01 mark

b. (i). Archemedes principle states that Any object partially or totally Immersed in a fluid experiences an upthrust which is equal to the weight of the fluid displaced by the body. 02 Marks

(ii). when a body is Immersed in liquid it exerts an upward force on a body. This force tends to reduce the weight of a body. @ 2 marks

(c). Weight in water (liquid) = 4N
Weight in air = 5.1 N
Relative density = ?
Density = ?

Relative density of object = $\frac{\text{Weight in air}}{\text{Loss in weight in liquid (upthrust)}}$ 0 of mark

$= \frac{5.1\ N}{(5.1-4)N}$ 00½ mark

$= \frac{5.1}{1.1}$.

Relative Density = 4.64 01mark.

Relative Density = $\frac{\text{Density of substance}}{\text{Density of water}}$ 00½ mark

$4.64 = \frac{\text{Density of Substance}}{1000 kg/m^3}$ 00½ mark

Density of Substance = $4.64 \times 1000 kg/m^3$

Density of object = $4,640 kg/m^3$ 0/mark

[TOTAL 10 Marks]

5 (a)(i) Adhesion – is the force of attraction between molecules of different substances. 0/mark while Coehesion is the force of attraction between Molecules of the same substance. 0/mark

(ii) Static equilbrium is one which object is not moving at all the condition 0/mark and Kinetic equilbrium is the equilbrium which the body is moving. 0/mark

(iii) Elasticity is the property of a body to return to its origional shape and size on removal of stretching force 01 mark

while

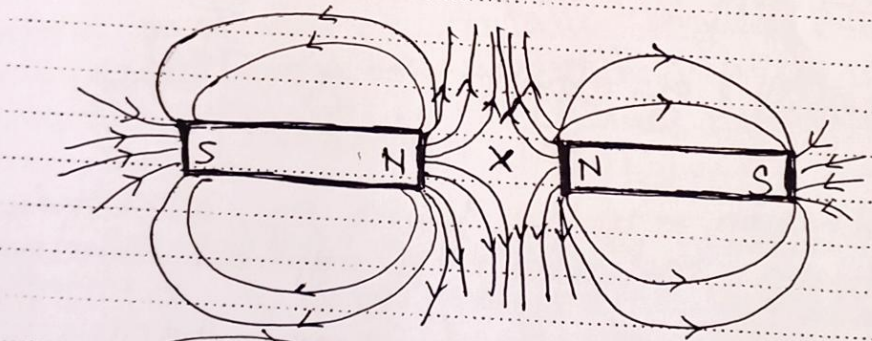
Viscosity is the force of friction which exists between the layers of liquids or gas. 01mark

(iv) Osmosis is the movement of Solvent molecules from low to high concentration region through semipermable membrine 01 mark

while

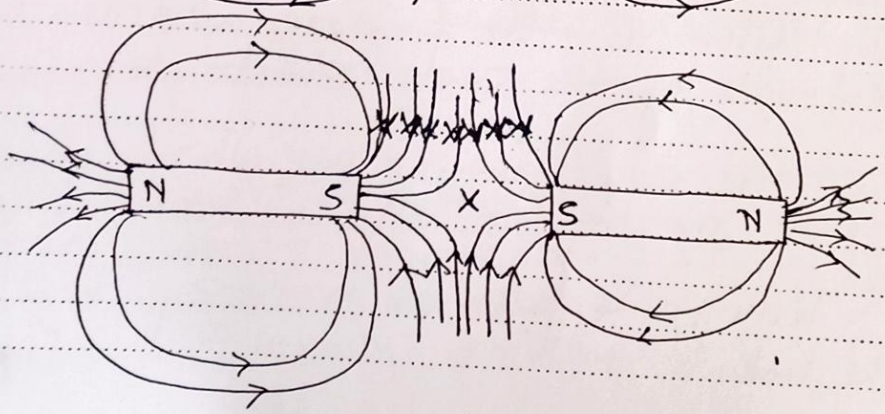
Diffusion is the movement of molecules/particles from high concentration region to low concentration region. 01 mark

(b). The Law states that "Like poles repeal, unlike poles attract"



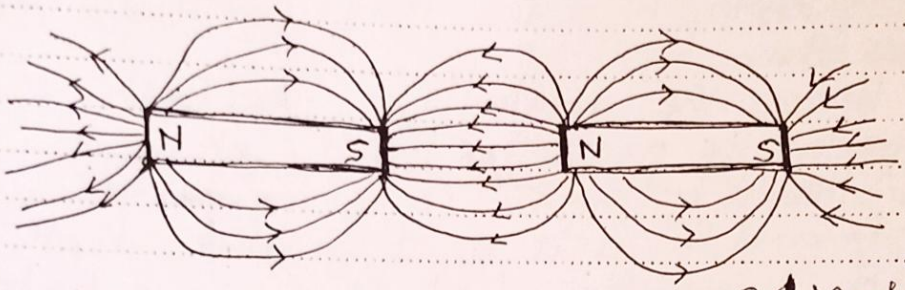
Like poles

01 mark



Like Poles

X - Neutral point.



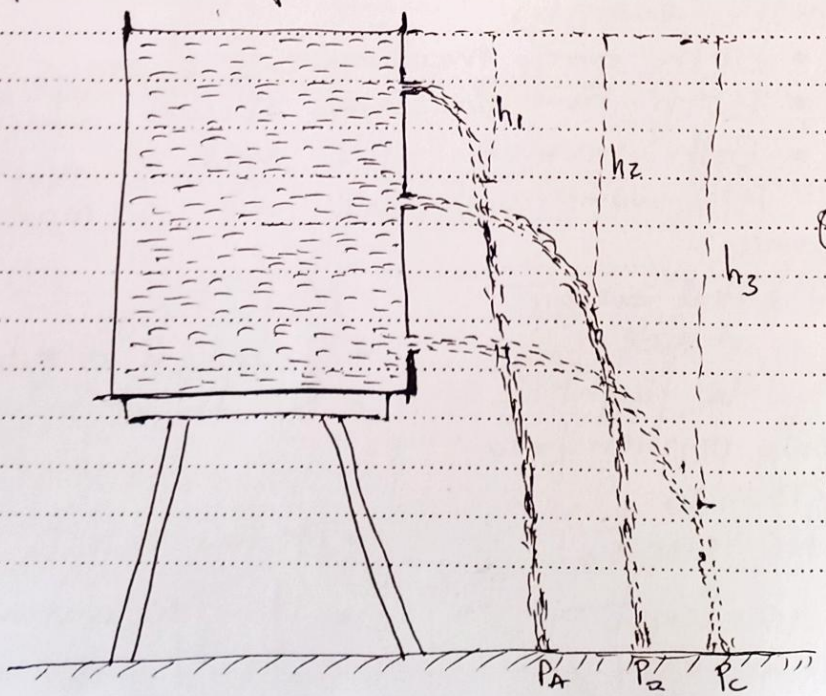
01 Marks.

Unlike poles

[TOTAL 10 Marks]

6. (a) A big elephant can walk on a muddy road while a human being can sink because of the wide area of the elephant feet ensures that the pressure due to its weight is low. On the other hand the small the area of a human feet may cause more pressure on the muddy and lead to sinking. 03 Marks

(b) (i) The diagram to demonstrate that liquid pressure depends on depth.



04 Marks.

$h_1 < h_2 < h_3$

(ii). The depth h_1 is less than depth h_2 and less than h_3.
Hence the greater/large the depth the greater the pressure of liquid.
Pressure of liquid P_C is greater than that at P_A and P_B as it is at greatest depth.
P_B is greater that P_A because P_B has greater depth as compared to P_A and P_B is less than P_A because its depth is less than that of P_C
ie the pressure depends on depth as there is thrust force pressing the liquid down as the height increases. 03 Marks

[TOTAL 10 Marks]

7. (a). Power — is the rate of doing work while the SI unit of power is watt, Joule/second 01½ marks
Energy - is the ability to do work. 01½ marks
The SI unit of energy is Joule 1½ Marks

(b). The similarities between Potential energy and Kinetic energy.
- Both are mechanical energy.
- Both have the same SI unit which is Joule(J)
- Both involves the mass of a body.

ANY TWO → @00½ mark ⇒ 01 Mark

Differences:

Potential energy	Kinetic energy
• Body is at state of Rest	• The body is at state of motion
• Involves the height of the body position from the ground.	• NO Height included
• Has NO velocity.	• It has velocity
• Has acceleration due to gravity	• NO Acceleration due to gravity

ANY 3 [03 Marks]

(c). Given: Mass = 90kg.
height = 10M.
time = 4sec.

Work done = force x distance = Mgh = Potential energy

∴ PE = mgh
= 90kg x 10m/s x 10m
= 9,000 Joules

0 1½ Marks

Power = $\frac{\text{work done}}{\text{time}}$

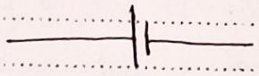
Power = $\frac{9,000}{4}$ J/s

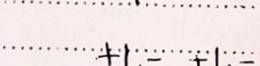
0 1½ Marks

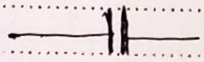
Power = 2250 watts

∴ The power generated by the Mass = 2250 watts.

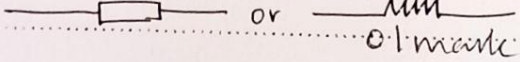
[TOTAL 10 Marks]

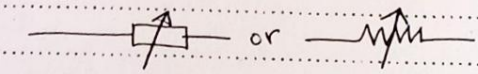
8. (a) (i). A cell:  0 1 mark

(ii). Battery:  0 1 mark

(iii). Capacitor:  0 1 mark

iv). Switch:  0 1 mark

(v). Resistor:  or 0 1 mark

vi) Rheostat.  or 0 1 mark

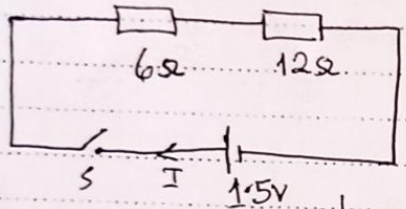
(b). Given:

Potential difference = 1.5V.

Resistance, $R_1 = 6\Omega$.

Resistance, $R_2 = 12\Omega$

Series



0 1 mark

$I = V/R_T$.

But $R_T = R_1 + R_2$

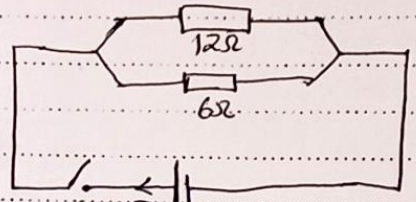
$= 6\Omega + 12\Omega$

$R_T = 18\Omega$

$\therefore I = \frac{1.5V}{18\Omega}$ 0 1 mark

$I = 0.08$ Ampere

Parallel



0 1 mark

$I = V/R_T$

But $R_T = \frac{R_1 \times R_2}{R_1 + R_2}$

$= \frac{6 \times 12}{6+12}\ \Omega$ 0 1 mark

$= \frac{72}{18}\ \Omega$

$R_T = 4\Omega$

$\therefore I = V/R_T = \frac{1.5}{4}$ Ampere

$= 0.375$ Ampere

[TOTAL 10 marks]

9.(a) Simple Machine – Is any device with a few or No moving parts that is used to Modify Motion and Magnitude of force in performing work. 02 marks

OR – Is any device that use applicati of a single force to simplify work.

(b) (i) Load is Mass moved by a machine or is the weight of an object beeing moved by a simple machine. 01 Mark

(ii) Effort is the force applied in moving the load.

or = The force applied to move a load. 01 mark

C) (i). The Efficiency of a machine is always less than 100%.

. This is because some of the Effort applied is used to overcome frictional force in the moving part of the system. Therefore the mechanical advantage is numerically always less than the velocity ratio. 02 Marks

(ii). Given data

velocity Ratio = 5

Work input = 600J.

load = 400 N

distance moved by load = 0.5M

Mechanical Advantage = ?

~~Velocity Ratio~~

Efficiency = ?

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The velocity Ratio = $\frac{\text{Effort distance}}{\text{load distance}}$,

$5 = \frac{\text{Effort distance}}{0.5\text{M}}$

Effort distance = (5×0.5)M 01 mark

Effort distance = 2.5M

Effort distance = 2.5M.

Now work Input = Effort x Effort distance

Effort = $\frac{\text{work input}}{\text{distance moved by Effort}}$

Effort = $\frac{600\text{J}}{2.5}$ 01 Mark

∴ Effort = 240N

Then Mechanical Advantage = $\frac{\text{load}}{\text{Effort}}$

= $\frac{400\text{N}}{240\text{N}}$ 01 Mark

M.A = 1.67

The Efficiency = $\frac{\text{MA}}{\text{V.R}} \times 100\%$

Efficiency = $\frac{1.67}{5} \times 100\%$

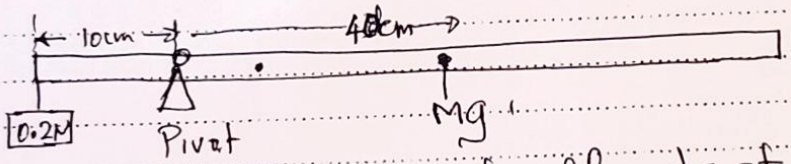
Efficiency = 33.4% 01 Mark

[TOTAL 10 Marks]

(ii). stable equilbrium is the state of an object to return to its origional/rest position after being displaced slightly. 01 mark

Unstable equilbrium is the state of an object which when slightly displaced fail to return to its rest position. 01 mark

(iii). Diagram:



The mass of a ruller m, should acts at the centre of the metre rule ie at 50cm Mark. 01 mark

Hence clockwise Moments = Anticlockwise moments 01 mark

$\therefore Mg \times 40cm = 0.2N \times 10\,cm$

$\therefore Mg = \frac{0.2N \times 10\,cm}{40cm}$ 01 mark

$Mg = \frac{2}{40} N$

$Mg = 0.05N = \frac{0.05}{10} kg$

$\therefore M = \frac{0.05N}{10 N/kg} = 0.005kg$ 01 marks

$\therefore$ The Mass of metre rule = 5g.

[TOTAL 15 Marks]